

MODULE - 5

Ktunotes.in

MOHAMMED FAHEEM T

DEPT. OF MECH. ENG .

KMCT COLLEGE OF ENGINEERING

Bio Mass Energy

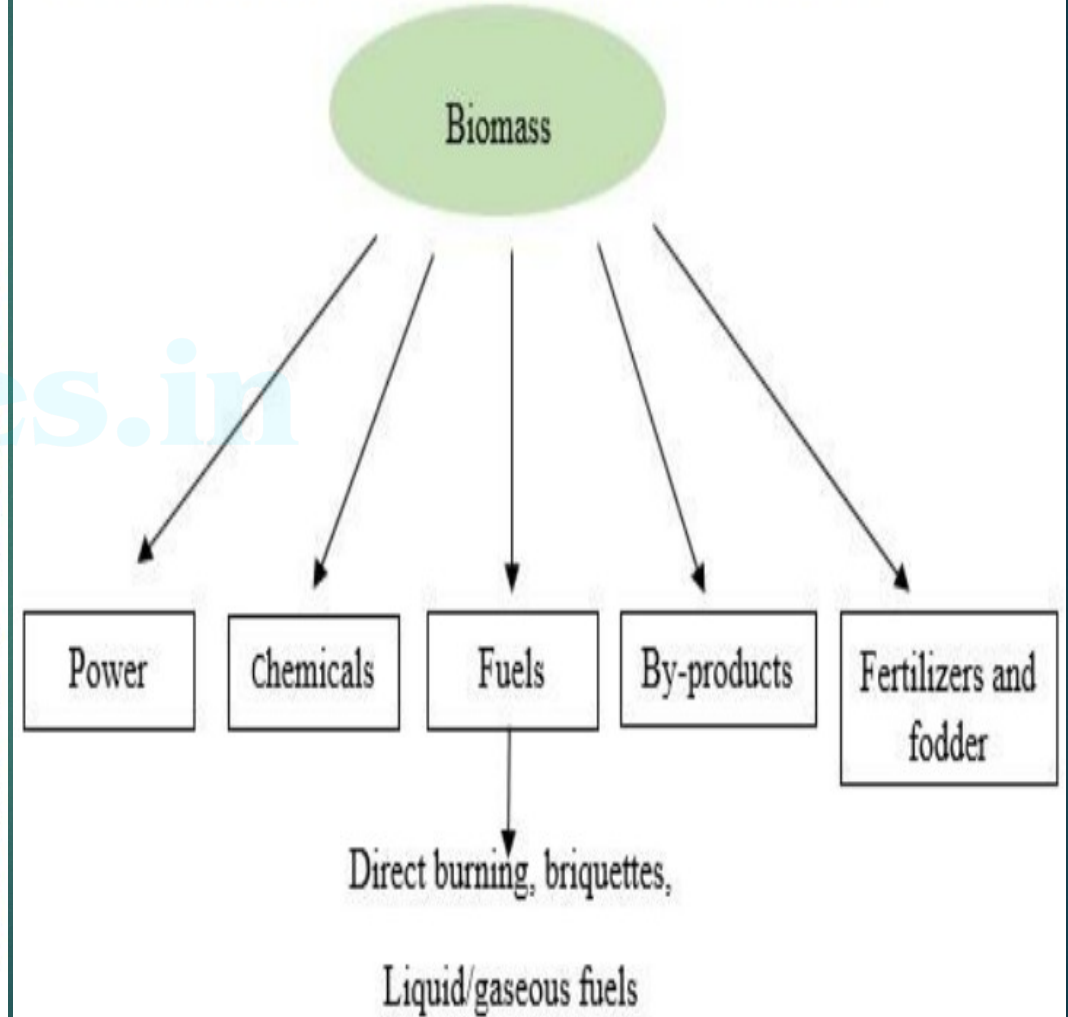
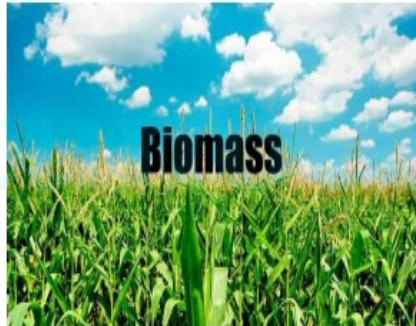
2

- Biomass – Composition and Characteristics :

Photosynthesis



In the process of photosynthesis, plants convert radiant energy from the sun into chemical energy in the form of glucose—or sugar.



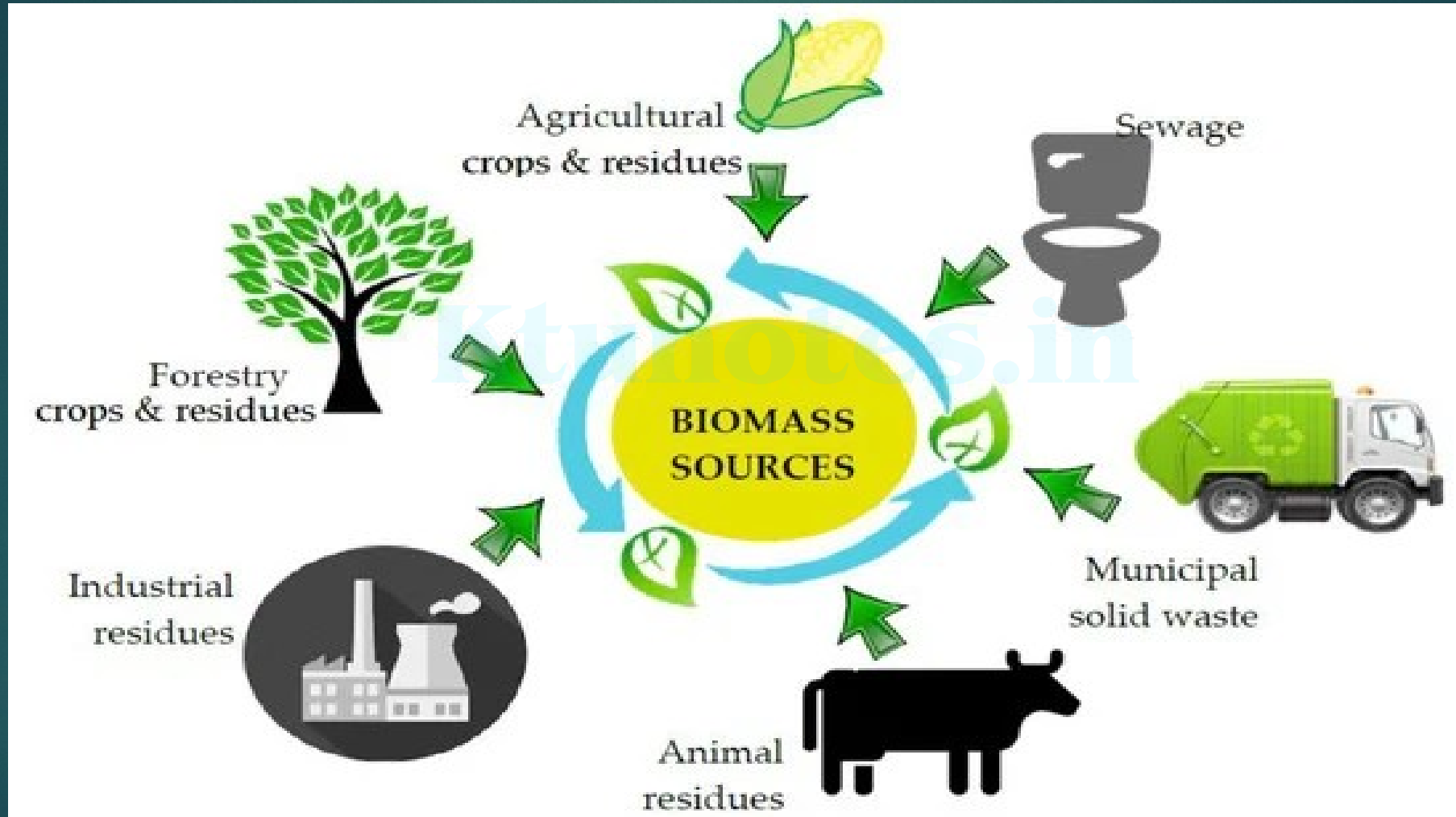
BIOMASS

3

- Biomass is an organic material that comes from living organisms, such as plants and animals.
- The most common biomass materials used for energy are plants, wood, and waste products.
- Biomass is one of the renewable energy sources.
- The energy from these organisms can be transformed into usable energy through direct and indirect means.
- Biomass can be burned to create heat (direct), or processed into biofuel (indirect).

BIOMASS SOURCES

4



Biomass-To-Energy Conversion Process

5

[1] Thermochemical Conversion Process

[2] Biochemical Conversion Process

ktunotes.in

THERMO CHEMICAL CONVERSION PROCESS

6

1) COMBUSTION

2) PYROLYSIS

3) GASIFICATION

Ktunotes.in

COMBUSTION

7

■ Biomass + Stoichiometric oxygen $\rightarrow$ Hot combustion products

- Combustion or burning is a complex sequence of exothermic chemical reactions between a fuel and an oxidant accompanied by the production of heat or both heat and light in the form of either a glow or flames, appearance of light flickering.
- In a complete combustion reaction, a compound reacts with an oxidizing element, such as oxygen. the
- products are compounds of each element in the fuel with the oxidizing element.



- Pyrolysis is the chemical decomposition of a condensed substance by heating.
- It does not involve reactions with oxygen or any other reagents but can take place in their presence
- Pyrolysis is a special case of thermolysis, and is most commonly used for organic materials.
- Extreme pyrolysis, which leaves only carbon as the residue, is called carbonization and is related to the chemical process of charring.
- Higher efficiency is achieved by the flash pyrolysis where finely divided feedstock is quickly heated to between 3500 and 500 0C for less than 2 seconds.

PYROLYSIS

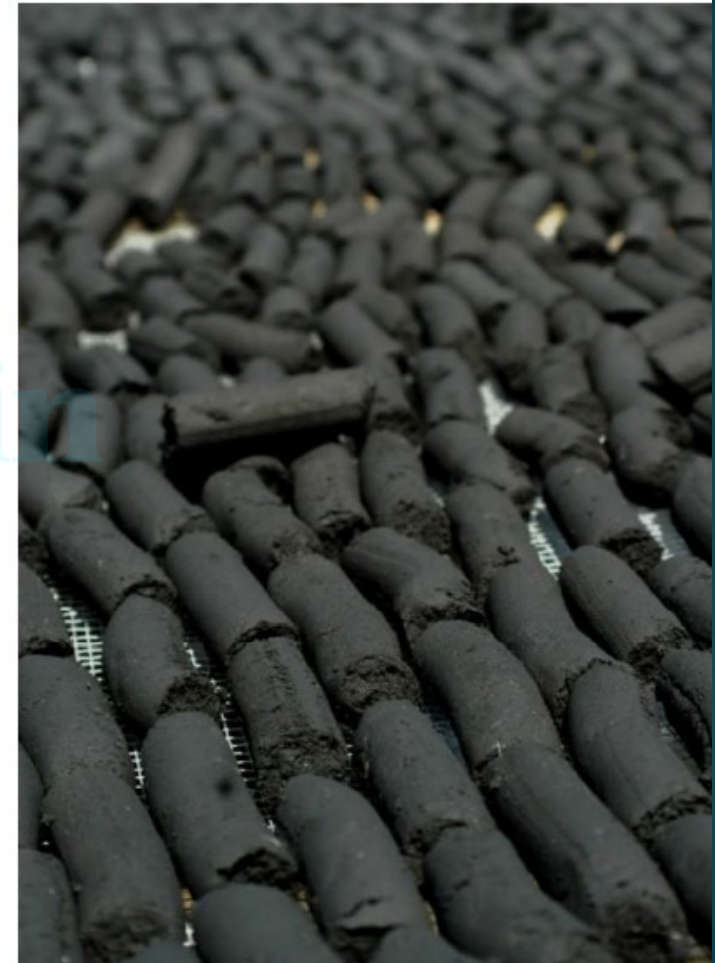
9



**Pyrolysis of
Biomass**

PYROLYSIS PRODUCTS

10



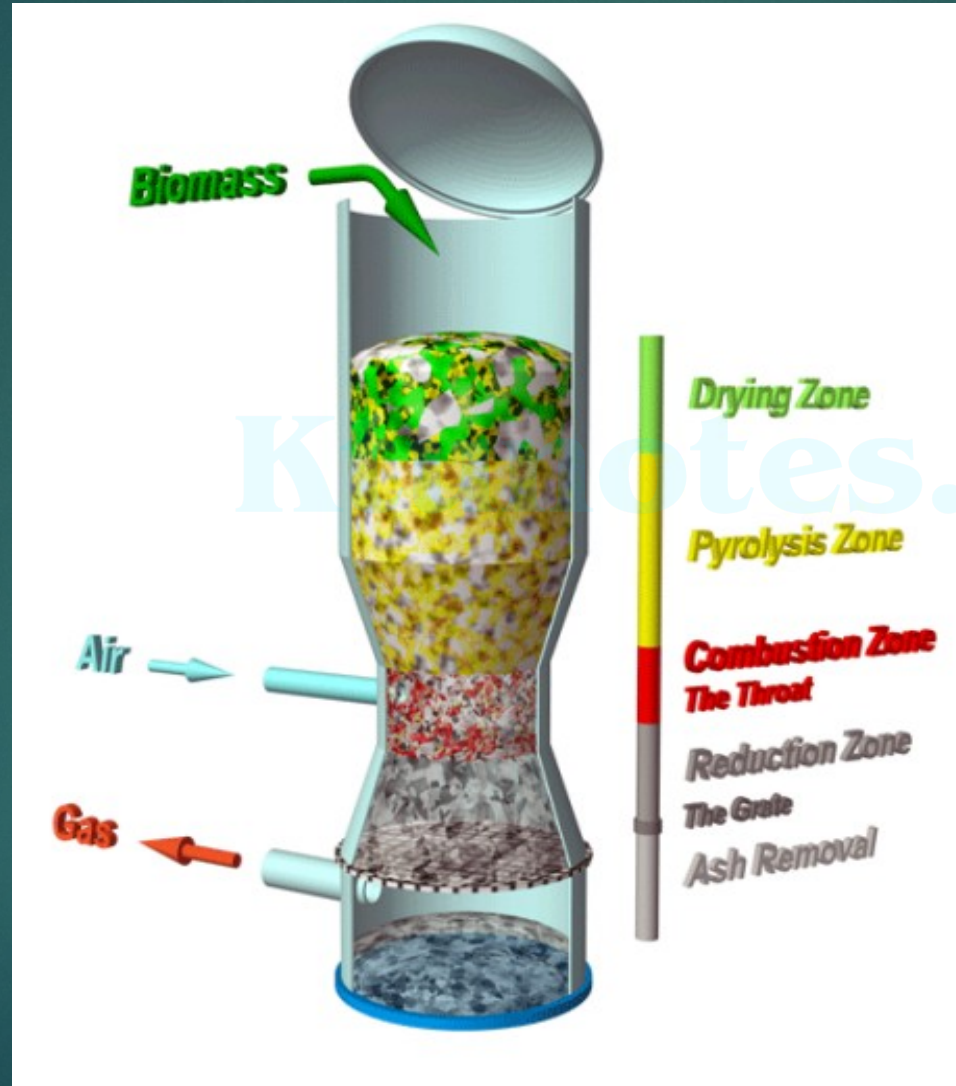
GASIFICATION

11

- Biomass + Limited oxygen $\rightarrow$ Fuel gas + Pyrolysis oils + Char + Ash + Steam
- Gasification is a process that converts carbonaceous materials, such as coal, petroleum, or biomass, into carbon monoxide and hydrogen by reacting the raw material at high temperatures with a controlled amount of oxygen and/or steam.
- The resulting gas mixture is called synthesis gas or syngas and is itself a fuel.
- Conversion of solid fuels into combustible gas mixture called producer gas or syngas($\text{CO} + \text{H}_2 + \text{CH}_4$)

GASIFICATION

12



THERMO CHEMICAL CONVERSION PROCESS

13

	Combustion	Gasification	Pyrolysis
Main products	heat, flue gas	combustible gas, heat	Oil, combustible gas & char
Carbon conversion	>99%	80-95%	≈ 75%
Main product constituents (raw gas)	CO ₂ , H ₂ O, N ₂	CO, H ₂ , CH ₄ , CO ₂ , H ₂ O, N ₂ , tar	oil, tar vapour, CO, H ₂ , CH ₄ , CO ₂ , H ₂ O, char
Oxygen stoichiometry (λ)	>1, typically 1.3 for solid fuels	$0 < \lambda < 1$, typically $0.2 < \lambda < 0.4$	0
Chemical reactivity of the main product	inert	combustible, but stable	combustible, reactive
Physical appearance	gas	gas	solid, liquid & gas
Heating value	0	typically 5 – 20 MJ m ⁻³	16-19 MJ kg ⁻¹

BIO CHEMICAL CONVERSION PROCESS

14

1) ANAEROBIC DIGESTION.

2) ALCOHOLIC FERMENTATION.

Ktunotes.in

ANAEROBIC DIGESTION.

15

- Method of production of biogas from waste biomass.
- The main products: biogas and innocuous digested sludge solids.
- Primary and secondary sludges are degraded in anaerobic digesters.
- Primary sludge consists of the settled solids from primary clarifiers and any colloidal wastes associated with the solids.
- Secondary sludge consists mostly of waste activated sludge or the humus from trickling filters.
- The mixture of primary and secondary sludges, which is an ideal medium for bacterial growth, contains 60% to 80% organic matter (dry weight) in the forms of carbohydrates, fats, and proteins.
- The sludges are rich in substrates (food) and nutrients and contain a large number and diversity of bacteria required for AD.

- Biogas is clean environment friendly fuel that can be obtained by anaerobic digestion of animal residues and domestic and farm wastes, abundantly available in the countryside.
- Biogas is an important renewable energy resource, especially for rural areas in India.
- Average calorific value of biogas is 20MJ/m³

Component	Composition
Methane(CH ₄)	55-65%
Carbon dioxide(CO ₂)	35-45%
Hydrogen sulphide(H ₂ S)	0.5-1.0%
Water vapor	Traces

Major types of Biogas plants:

1) Floating drum type

2) Fixed dome type

ktunotes.in

Floating drum type Biogas plant

18

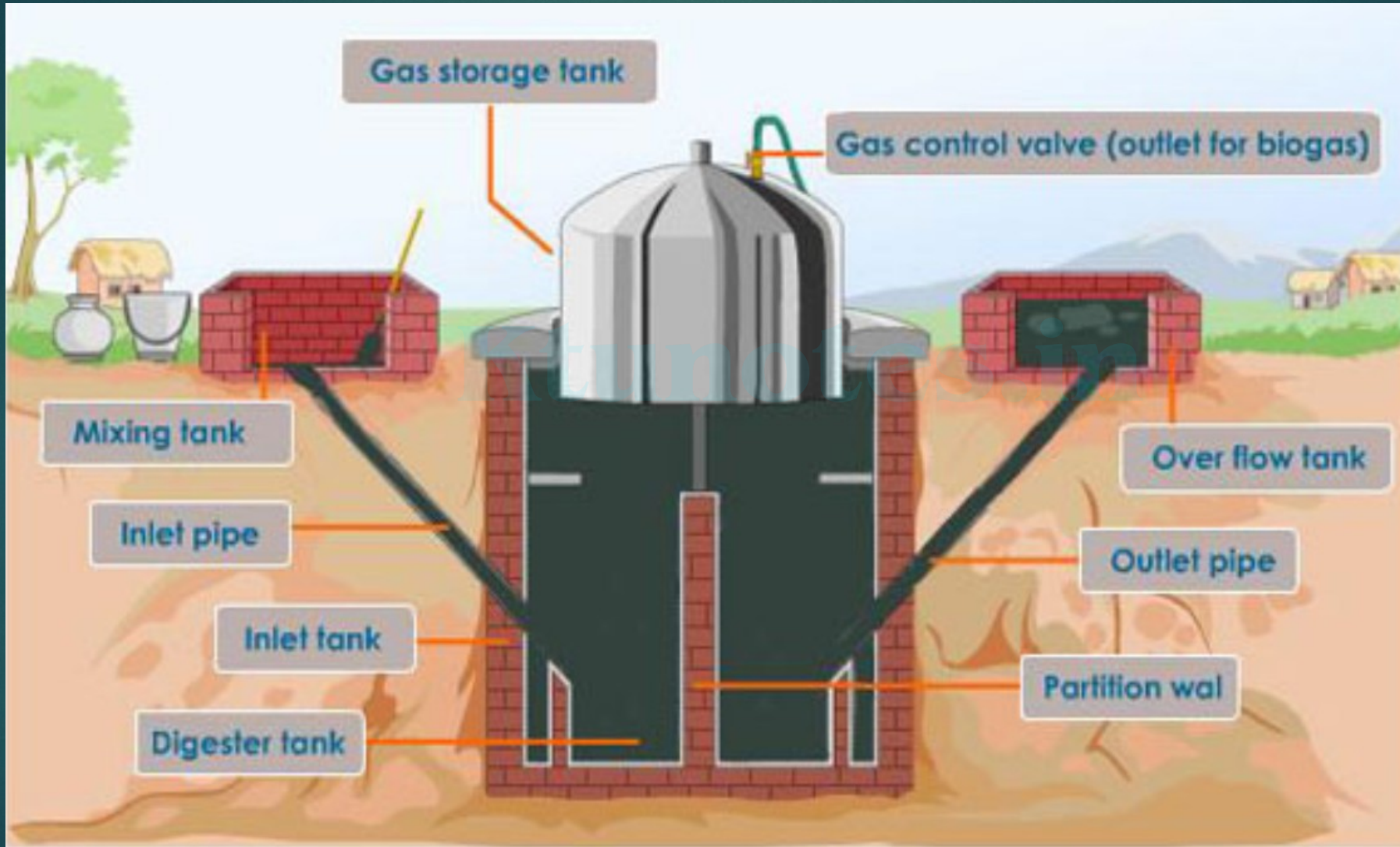
- **KVIC type biogas plant:**
- This mainly consists of a digester or pit for fermentation and a floating drum for the collection of gas.
- Digester is 3.5-6.5 m in depth and 1.2 to 1.6 m in diameter.
- There is a partition wall in the center, which divides the digester vertically and submerges in the slurry when it is full.
- The digester is connected to the inlet and outlet by two pipes.
- Through the inlet, the dung is mixed with water (4:5) and loaded into the digester.
- The fermented material will flow out through outlet pipe.
- The outlet is generally connected to a compost pit.

- In the first stage, the complex, organic substances contained in the waste are acted upon by a certain kind of bacteria, called acid formers and broken up into small-chain simple acids.
- In the second stage, these acids are acted upon by another kind of bacteria, called methane formers and produce methane and carbon dioxide.

Ktunotes.in

CONSTRUCTION

20



CONSTRUCTION

21

- The gas holder is a drum constructed of mild steel sheets.
- This is cylindrical concave shape.
- The top is supported radially with angular iron.
- The holder fits into the digester like a stopper. It sinks into the slurry due to its own weight and rests upon the ring constructed for this purpose.
- When gas is generated the holder rises and floats freely on the surface of slurry. A central guide pipe is provided to prevent the holder from tilting.
- The holder also acts as a seal for the gas.
- The cost of drum is about 40% of total cost of plant. It requires periodical maintenance. The unit cost of KVIC model with a capacity of 2 m³/day costs approximately Rs.14, 000 .

Economic Analysis of Renewable Energy Systems

22

- Renewable energy resources such as wind and solar are playing increasingly important role in present and future energy scenarios of both developed and developing countries.
- Decisions making on design and implementation of these renewable energy systems, require knowledge of economic and life cycle assessments of these systems in order to make an informed decision.
- The decisions should be based on : understanding the importance of economics ,conduct economic analyses such as life cycle cost, net present value, and payback period to assess economic feasibility of engineering systems.
- Environmental impact and sustainability as well as tax credit and incentives also play key roles in conducting an economic analysis of renewable energy systems.

Life Cycle Cost

23

- Life Cycle Cost (LCC) is the total cost of an item during its lifetime. It is the sum of all costs – both recurring and non-recurring – over the life of a product
- Life-cycle cost analysis (LCCA) is used to analyze the life cycle cost of a project.
- A LCCA takes into account all costs, from the cost of **construction**, cost of **fuel or repairs** to the **cost imposed by emissions** from the project. The analysis takes into account all known and projected costs.
- Costs such as the social cost from pollution are difficult to quantify and therefore the LCCA is not exact.
- The LCCA of a project allows firms to compare different projects to determine which is the cheapest in the long term.

Payback time

- Payback Period is nothing but the number of years it takes to recover the initial cash outlay invested in a particular project.
- Payback period refers to the amount of time it takes to recover the cost of an investment.
- People and corporations mainly invest their money to get paid back, which is why the payback period is so important.
- The shorter payback an investment has, the more attractive it becomes.